

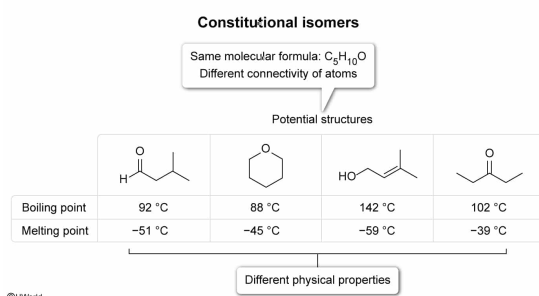
Which of the following characteristics is(are) shared by constitutional isomers?

- I. Same molecular formula
 - II. Nonsuperimposable mirror images
 - III. Different connectivity of atoms
 - IV. Usually have different physical properties
- ☐ A. I only [5%]
☐ B. I and III only [23%]
☐ C. II and IV only [1%]
☒ D. I, III, and IV only [69%]

Correct

68% answered correctly

Explanation:



Isomers are molecules with the same molecular formula, and they are **classified** based on the connectivity (ie, constitutional isomers) and spatial orientation (eg, stereoisomers) of the atoms in their molecular structures. **Constitutional isomers**, also known as **structural isomers**, are molecules that contain the same **molecular formula (Number I)** and **differ in the order of atom connectivity** at one or more points within the structure (**Number III**). Because the atoms in constitutional isomers are connected differently, constitutional isomers are different molecules. Therefore, constitutional isomers usually have **different physical properties** (eg, boiling point, melting point) (**Number IV**).

Stereoisomers are molecules that have the *same* molecular formula and atom connectivity but differ in spatial orientation. Types of stereoisomers include **diastereomers**, **geometric isomers**, and **enantiomers**, which are nonsuperimposable mirror images (**Number II**).

Educational objective:

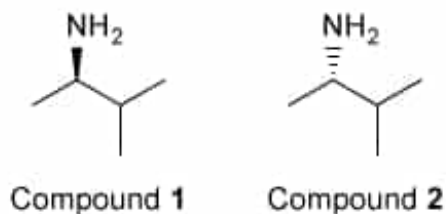
Isomers are molecules with the same molecular formula, and they are classified based on atom connectivity and spatial orientation. Constitutional isomers (structural isomers) are molecules with the same molecular formula that differ in the order of atom connectivity and have different physical properties.

Time spent: 0 sec

Subject:
Organic Chemistry

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System:
Introduction to Organic Chemistry



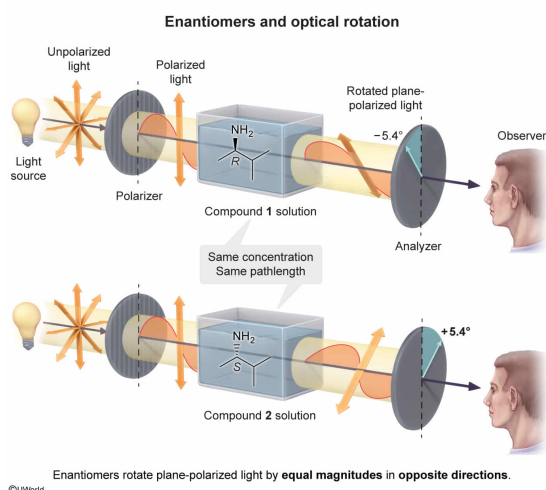
If Compound 1 has a specific rotation of $-5.2\ (^{\circ}\cdot\text{mL})/(\text{g}\cdot\text{dm})$, the specific rotation of Compound 2 must be:

- ☐ A. $-5.2\ (^{\circ}\cdot\text{mL})/(\text{g}\cdot\text{dm})$ [3%]
- ☐ B. $0\ (^{\circ}\cdot\text{mL})/(\text{g}\cdot\text{dm})$
- ☒ C. $+5.2\ (^{\circ}\cdot\text{mL})/(\text{g}\cdot\text{dm})$ [94%]
- ☐ D. $+10.4\ (^{\circ}\cdot\text{mL})/(\text{g}\cdot\text{dm})$

Correct

94% answered correctly

Explanation:



Specific rotation is the degree to which **chiral molecules** rotate **plane-polarized light**, and it is unique to each chiral molecule. Clockwise rotations are designated as (+) whereas counterclockwise rotations are (-). The **specific rotation** is a constant and describes the direction (+ or -) and magnitude (number of degrees) of a solution of normalized concentration (1 g/mL) and pathlength (1 dm).

Isomers are molecules that have the same molecular formula but have either different connectivity (constitutional isomers) or spatial orientation (stereoisomers). **Enantiomers** are nonsuperimposable mirror images where each corresponding **stereocenter** is opposite in configuration. The specific rotations of enantiomers are **equal in magnitude** but **opposite in direction (sign)**.

In this question, Compounds 1 and 2 are stereoisomers, where the **configuration** of the stereocenter in Compound 1 is *R* and the configuration of the stereocenter in Compound 2 is *S*. Consequently, Compounds 1 and 2 are enantiomers, and the specific rotations of Compounds 1 and 2 are equal in magnitude and opposite in direction. The question states

that the specific rotation of Compound **1** is $-5.2 (^{\circ}\cdot\text{mL})/(\text{g}\cdot\text{dm})$; therefore, the **specific rotation** of **Compound 2** must be **$+5.2 (^{\circ}\cdot\text{mL})/(\text{g}\cdot\text{dm})$** .

(Choice A) Although enantiomers have specific rotations of the same magnitude, the direction the plane-polarized light rotates is *opposite in direction*. Because Compound **1** and Compound **2** are enantiomers, they *cannot* have the same specific rotation.

(Choice B) Because Compounds **1** and **2** are enantiomers, a 50:50 mixture of these compounds is a racemic mixture. Racemic mixtures are optically inactive and have an *observed* rotation of 0° .

(Choice D) Although the specific rotation of $+10.4 (^{\circ}\cdot\text{mL})/(\text{g}\cdot\text{dm})$ is opposite in sign versus Compound **1**, it is *not* equal in magnitude. Therefore, this value *cannot* be the specific rotation of Compound **2**, the enantiomer of Compound **1**.

Educational objective:

Specific rotation is the normalized degree to which chiral molecules rotate plane-polarized light. Enantiomers each rotate plane-polarized light by the same magnitude but in opposite directions.

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